SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Project Name: Real-Time Stock & Stock Forecast Dashboard

1.0 Title of the Project

Real-Time Stock & Stock Forecast Dashboard

2.0 Introduction

2.1 Purpose

The purpose of this project is to develop a real-time stock monitoring and forecasting dashboard using Streamlit, Yahoo Finance API, and Prophet. This system will allow users to track stock prices in real-time, visualize historical trends, apply technical indicators, and predict future stock movements.

2.2 Intended Audience and Reading Suggestions

This document is intended for:

- Big Data Analysts
- Financial Analysts & Investors
- Developers & Data Scientists
- C-DAC Faculty & Project Evaluators

2.3 Project/Product Scope

- Real-time stock price tracking using Yahoo Finance API.
- Candlestick and line chart visualizations.
- Technical indicators: SMA, EMA, RSI.
- Stock forecasting using Prophet.
- User-friendly dashboar implemented with Streamlit.

2.4 References

- Facebook Prophet Documentation - https://facebook.github.io/prophet/
- Yahoo Finance API - https://pypi.org/project/yfinance/

3.0 Overall Description

3.1 Project/Product Perspective

This project aims to bridge the gap between real-time financial data analysis and predictive modeling, providing investors with a **data-driven decision-making tool**

3.2 Product Functions

- Fetch real-time and historical stock data.
- Display stock price movements with **interactive charts**.
- Compute **SMA, EMA, and RSI indicators**.
- Forecast future stock prices for the next **3-10 days**.

3.3 Operating Environment

3.3.1 Server

- **OS:** Linux/Windows
- **Application Framework:** Streamlit
- **Language:** Python
- **Database:** None (Real-time API-based data fetching)
- **Tool:** Prophet, Pandas, Plotly, yFinance

3.3.2 Client

- **Browsers:** Chrome, Firefox, Edge
- **Devices:** Desktop, Laptop, Mobile

3.4 Design and Implementation Constraints

- Limited to stock data available via Yahoo Finance API.
- Forecasting accuracy depends on historical data trends.

3.5 User Documentation

- Installation Manual.
- User Guide for dashboard navigation.

4.0 External Interface Requirements

4.1 User Interfaces

- **Interactive charts** for stock analysis.
- **Dropdown selection** for company stocks.
- **Sidebar options** for forecast and technical indicators.

4.2 Hardware Interfaces

Item	Specification
Processor	i3(11 Gen)
RAM	8 GB

Storage	256 GB (SSD)
OS	Linux/Windows
Network	Gigabit Ethernet/Wi-Fi
GPU	Nvidia RTX 2 GB

4.3 Communication Interfaces

- **Automated alerts via Email (Future Scope).**

5.0 System Features

5.1 Real-Time Stock Data Fetching

- Fetch **live stock data** from Yahoo Finance.
- Update prices every **60 seconds**.

5.2 Technical Indicators Computation

- Compute **Simple Moving Average (SMA)**.
- Compute **Exponential Moving Average (EMA)**.
- Compute **Relative Strength Index (RSI)**.

5.3 Stock Price Forecasting

- Predict **future stock prices** using Prophet.
- Display **3, 5, or 10-day** forecast.
- Graphical representation of forecasted prices.

6.0 Non-Functional Requirements

6.1 Performance Requirements

- Dashboard response time should be **<2 seconds**.
- Forecast calculations should complete within **5 seconds**.

6.2 Security Requirements

- HTTPS protocol for data requests.
- Secure API calls to prevent unauthorized data access.

6.3 Software Quality Attributes

- **Reliability:** System should handle API failures gracefully.
- **Maintainability:** Code should be modular and well-documented.

7.0 Acceptance Criteria

- Must fetch stock data in real-time.
- Must display technical indicators correctly.
- Must provide stock price predictions using Prophet.
- Dashboard should be accessible via web browsers.

8.0 Deliverables

- Fully functional **Streamlit-based dashboard**.
- Complete **Python source code**.
- User documentation with installation steps.
- Final project report.